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IMPERIAL

Predicting trajectories of acute illness using RNA velocity of whole blood

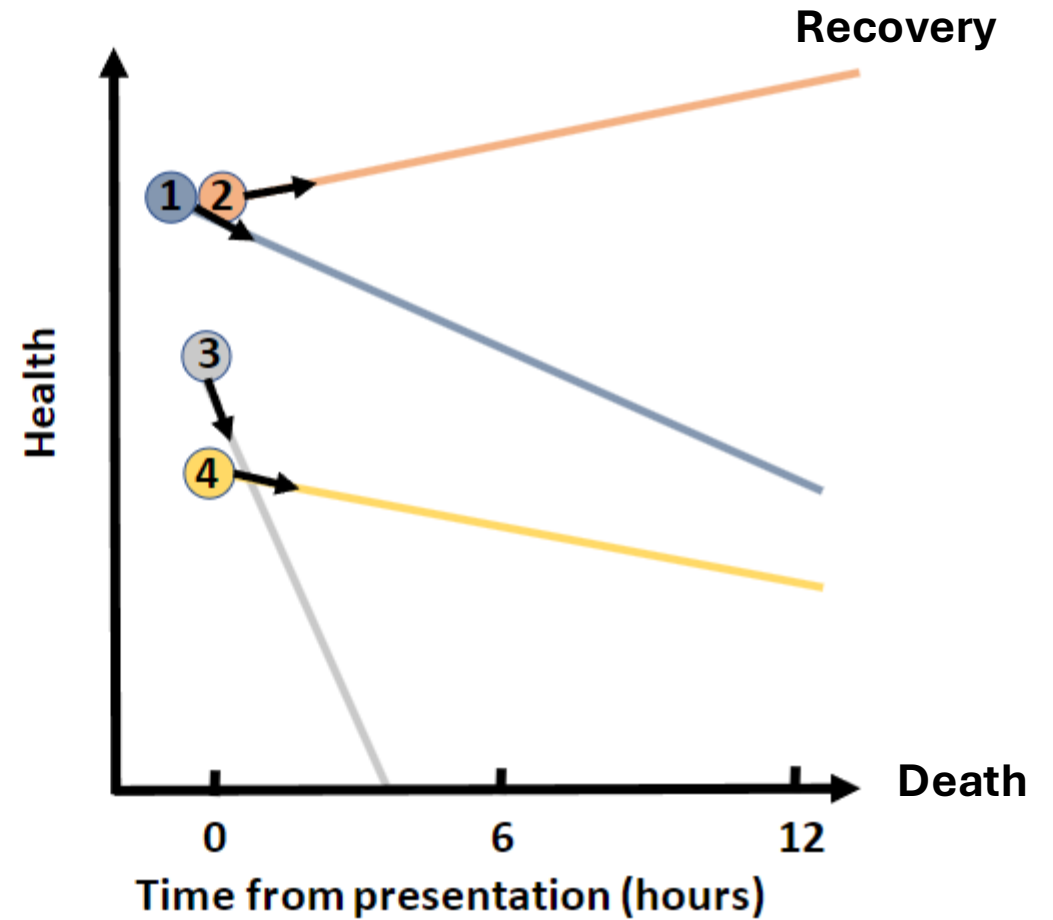
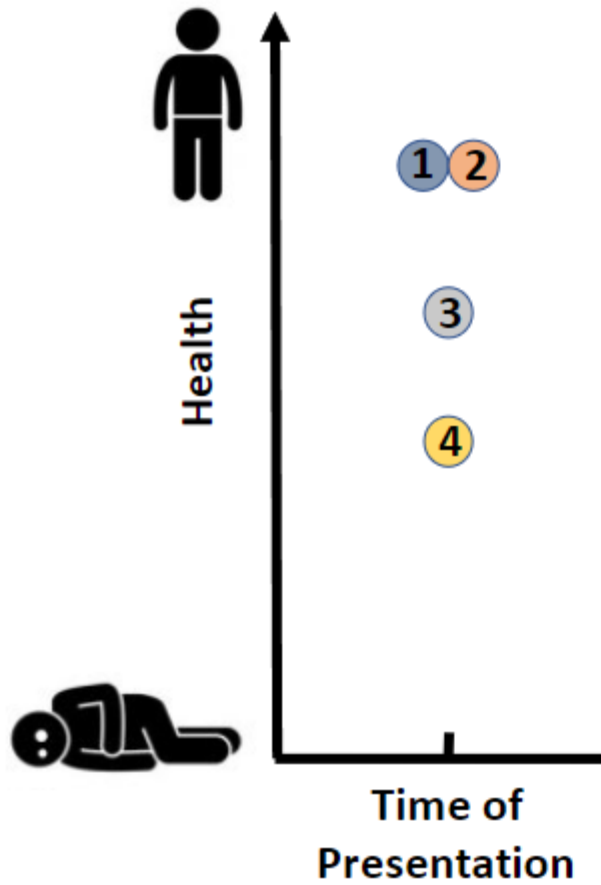
a.k.a. The last ~~7~~8 years of my life

Bioinformatics Webinar, 3rd March 2026

Dr Claire Dunican

Bioinformatician, Pediatric Infectious Disease

The problem: disease trajectory variation



Our Solution: develop a method
to predict future disease
trajectories

Background: RNA Velocity

nature

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Letter | Published: 08 August 2018

RNA velocity of single cells

[Gioele La Manno](#), [Ruslan Soldatov](#), [Amit Zeisel](#), [Emelie Braun](#), [Hannah Hochgerner](#), [Viktor Petukhov](#), [Katja Lidschreiber](#), [Maria E. Kastriti](#), [Peter Lönnerberg](#), [Alessandro Furlan](#), [Jean Fan](#), [Lars E. Borm](#), [Zehua Liu](#), [David van Bruggen](#), [Jimin Guo](#), [Xiaoling He](#), [Roger Barker](#), [Erik Sundström](#), [Gonçalo Castelo-Branco](#), [Patrick Cramer](#), [Igor Adameyko](#), [Sten Linnarsson](#)  & [Peter V. Kharchenko](#) 

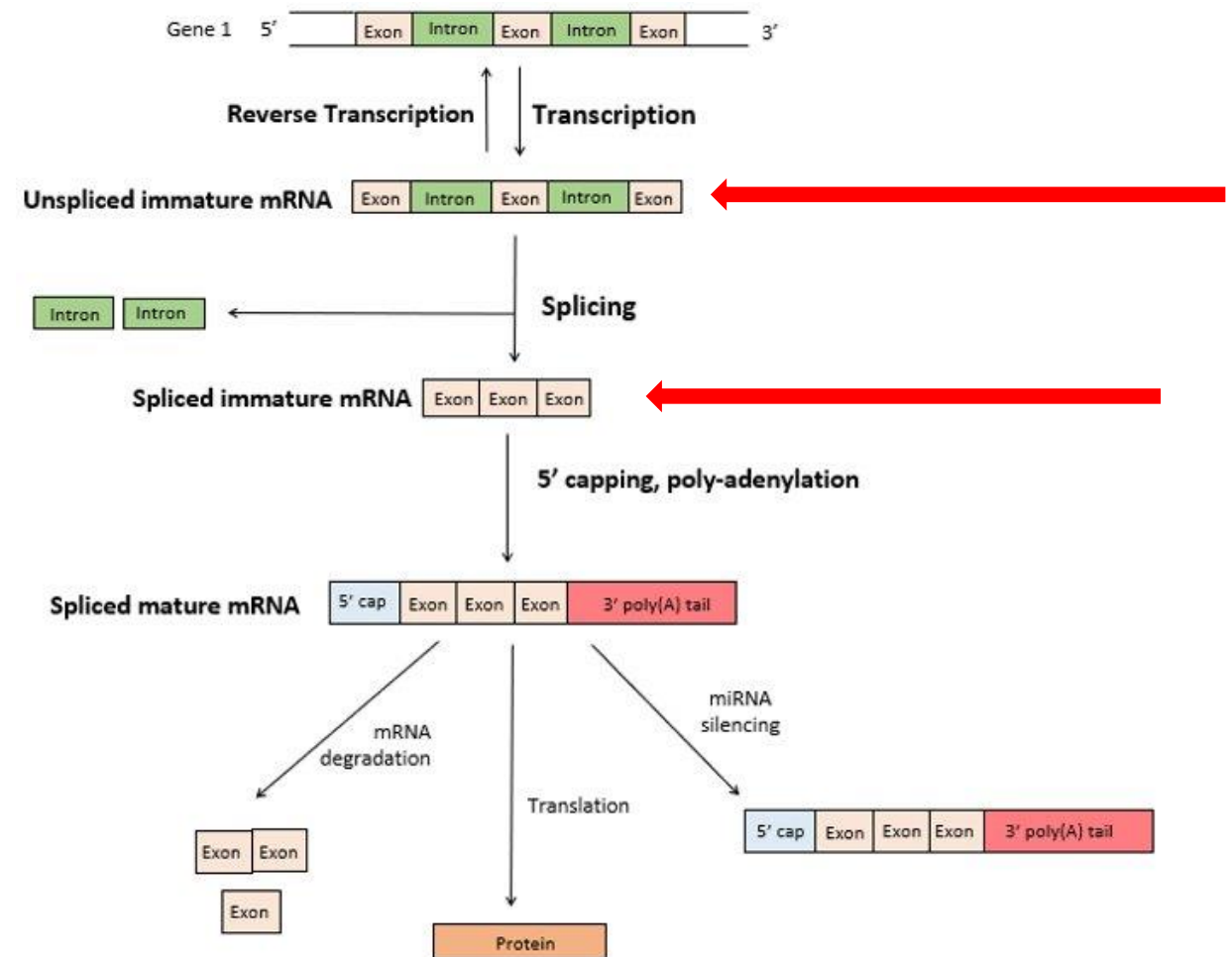
[Nature](#) **560**, 494–498 (2018) | [Cite this article](#)

238k Accesses | **1904** Citations | **671** Altmetric | [Metrics](#)

Abstract

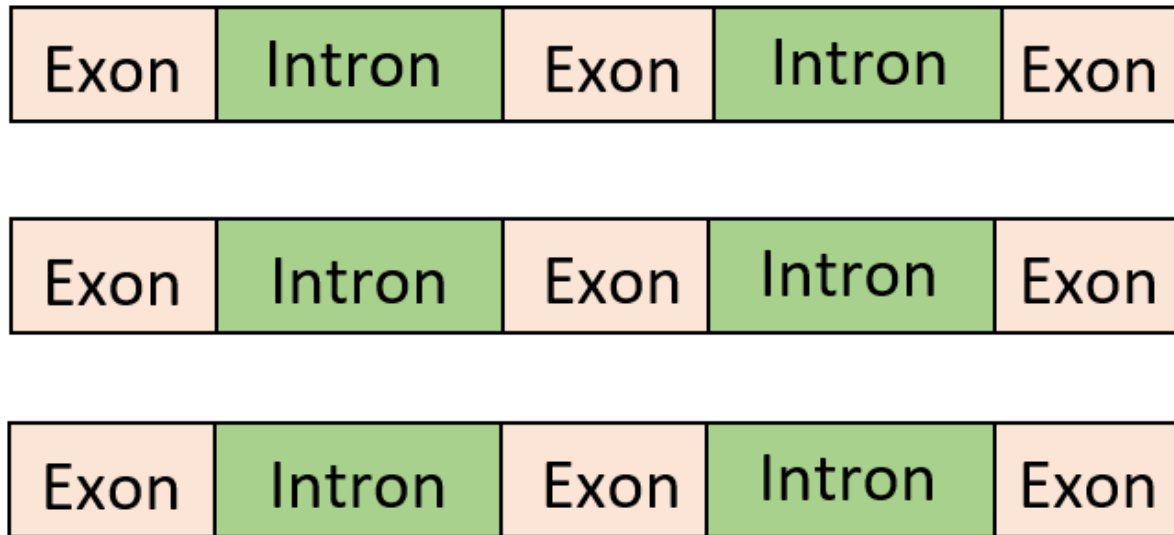
RNA abundance is a powerful indicator of the state of individual cells. Single-cell RNA sequencing can reveal RNA abundance with high quantitative accuracy, sensitivity and throughput¹. However, this approach captures only a static snapshot at a point in time, posing a challenge for the analysis of time-resolved phenomena such as embryogenesis or tissue regeneration. Here we show that RNA velocity—the time derivative of the gene expression state—can be directly estimated by distinguishing between unspliced and spliced mRNAs in common single-cell RNA sequencing protocols. RNA velocity is a high-dimensional vector that predicts the future state of individual cells on a timescale of hours. We validate its accuracy in the neural crest lineage, demonstrate its use on multiple published datasets and

Getting more out of transcriptomics

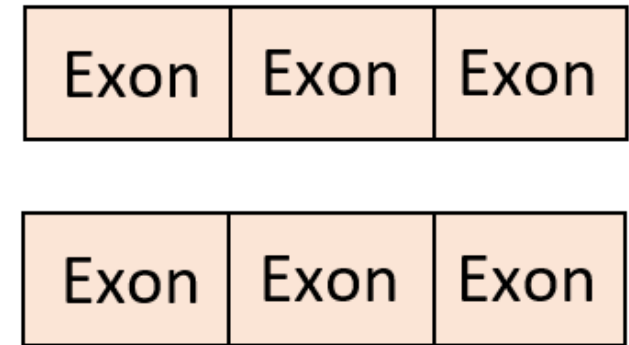


Background: RNA Velocity

Unspliced immature mRNA

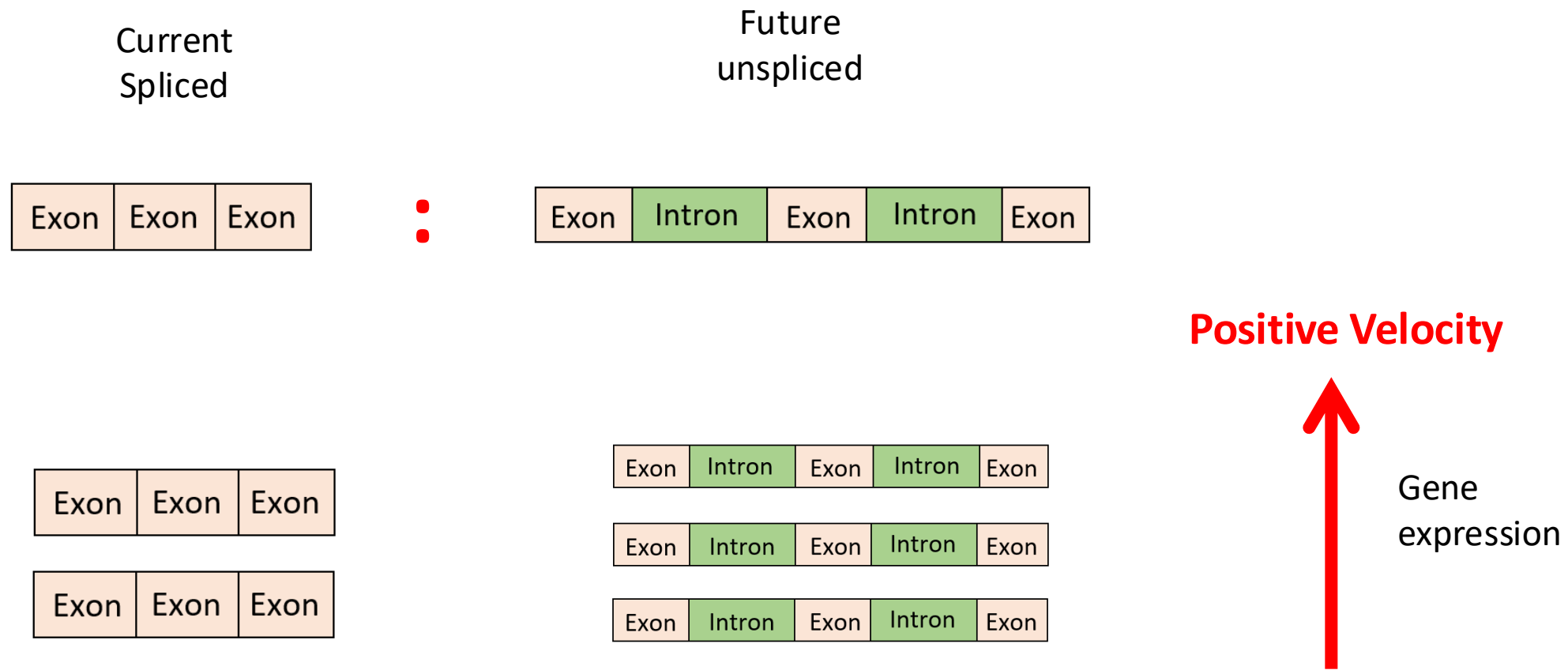


Spliced mRNA

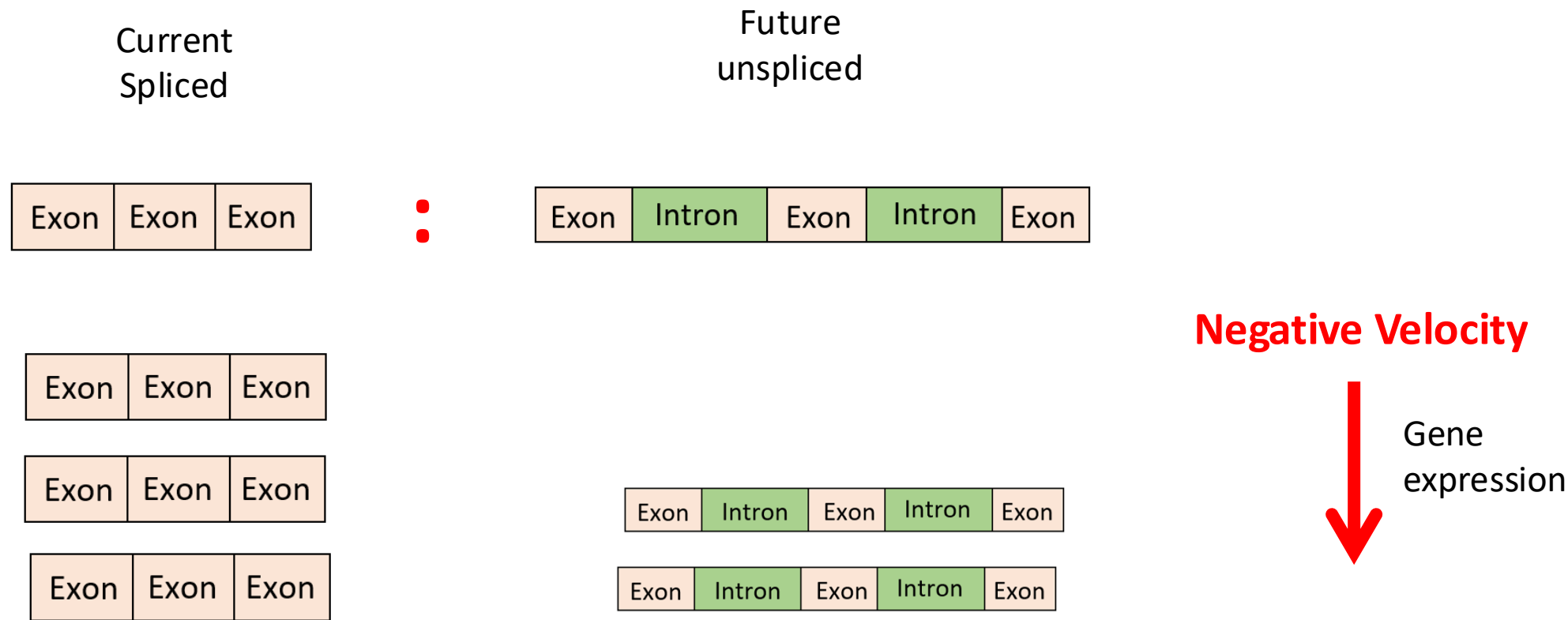


(What we normally refer to when talking about mRNA)

Background: RNA Velocity

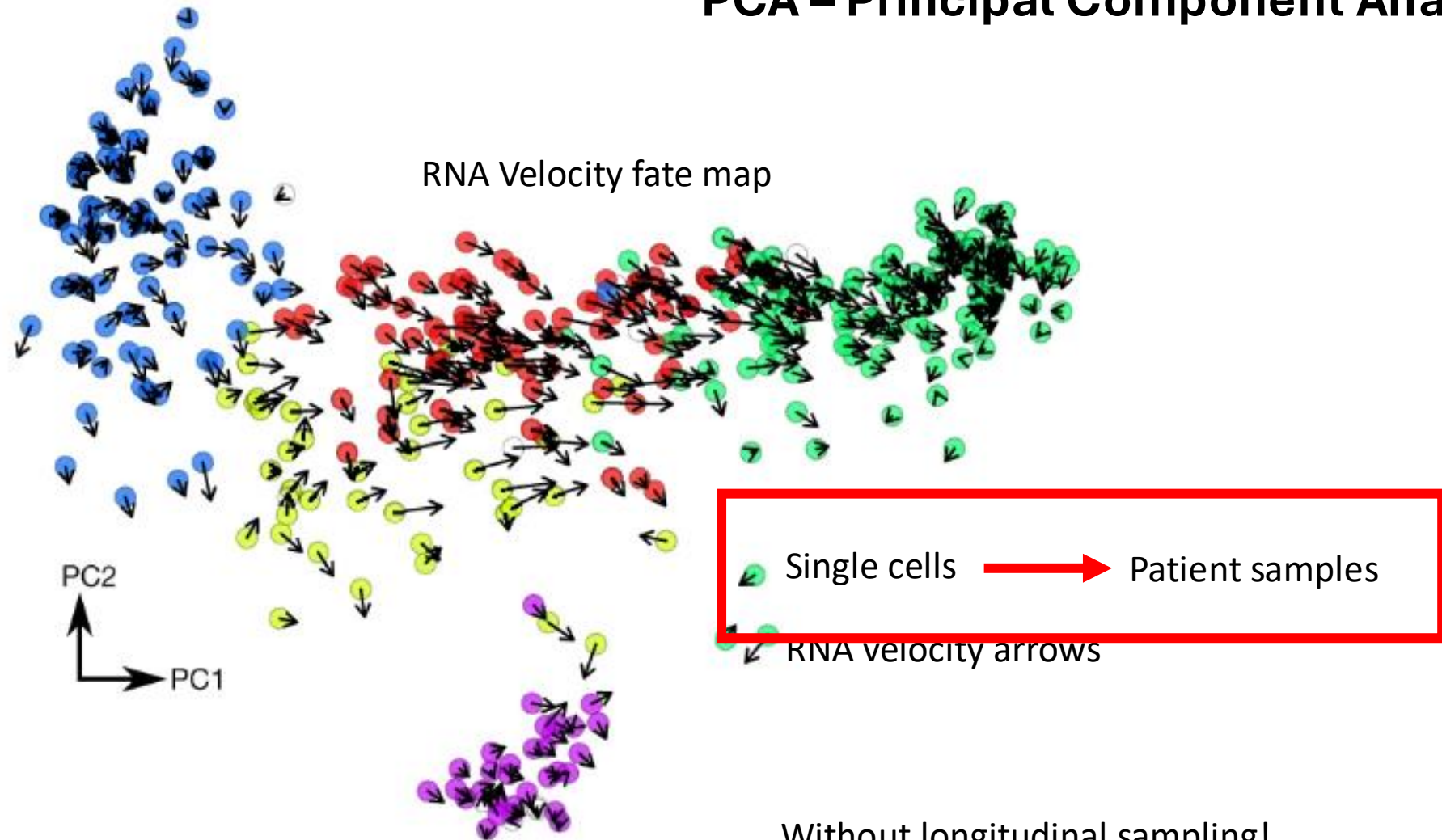


Background: RNA Velocity



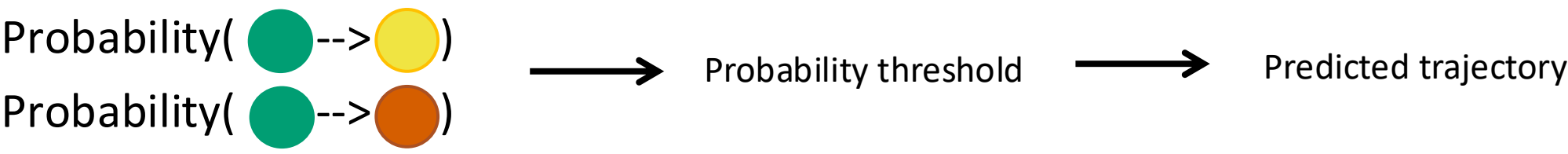
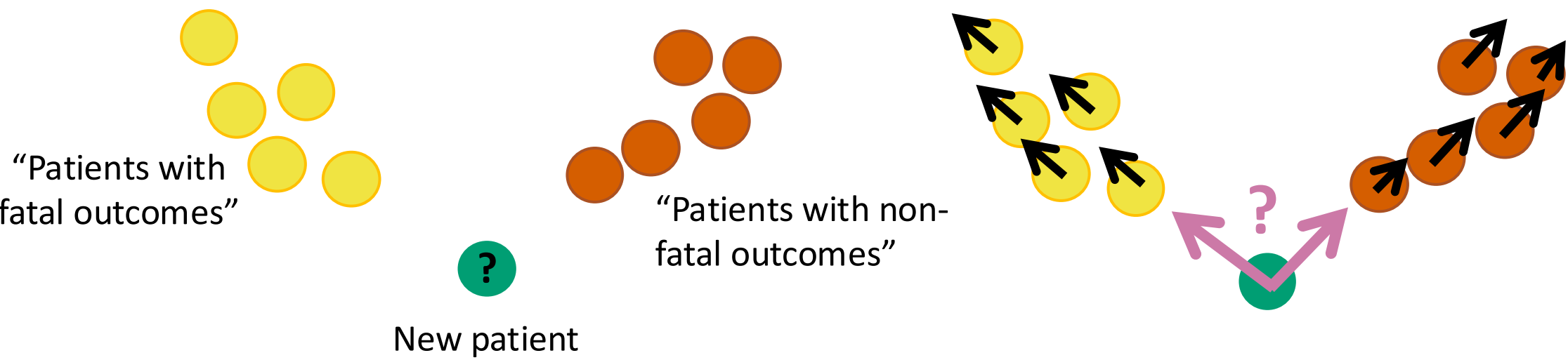
Background: RNA Velocity

PCA – Principal Component Analysis



VeloCD

A theoretical example



Assumption 1: current disease states == current transcriptomic state

Assumption 2: future disease trajectories == future transcriptomic states

Differentiation of COVID-19 from other emergency infectious disease

> Med. 2023 Sep 8;4(9):635-654.e5. doi: 10.1016/j.medj.2023.06.007. Epub 2023 Aug 18.

Diagnosis of childhood febrile illness using a multi-class blood RNA molecular signature

Dominic Habgood-Coote¹, Clare Wilson¹, Chisato Shimizu², Anouk M Barendregt³, Ria Philipsen⁴, Rachel Galassini¹, Irene Rivero Calle⁵, Lesley Workman⁶, Philipp K A Agyeman⁷, Gerben Ferwerda⁴, Suzanne T Anderson⁸, J Merlijn van den Berg³, Marieke Emonts⁹, Enitan D Carrol¹⁰, Colin G Fink¹¹, Ronald de Groot⁴, Martin L Hibberd¹², John Kanegaye², Mark P Nicol¹³, Stéphane Paulus¹⁴, Andrew J Pollard¹⁵, Antonio Salas¹⁶, Fatou Secka⁸, Luregn J Schlapbach¹⁷, Adriana H Tremoulet², Michael Walther⁸, Werner Zenz¹⁸; Pediatric Emergency Medicine Kawasaki Disease Research Group (PEMKDRG); UK Kawasaki Genetics consortium; GENDRES consortium; EUCLIDS consortium; PERFORM consortium; Michiel Van der Flier¹⁹, Heather J Zar⁶, Taco Kuijpers²⁰, Jane C Burns², Federico Martín-Torres⁵, Victoria J Wright¹, Lachlan J M Coin²¹, Aubrey J Cunningham¹, Jethro A Herberg¹, Michael Levin¹, Myrsini Kaforou²²

Affiliations + expand

PMID: 37597512 DOI: 10.1016/j.medj.2023.06.007

Free article

Perspective | Published: 17 October 2018

The value of transcriptomics of the immune response as a translational tool to distinguish

Akul Singhania, Robert

Nature Immunology

ORIGINAL RESEARCH article

6642 | Front. Cell Dev. Biol. 22 August 2014

Sec. Systems Biology Archive

Volume 2 - 2014 | https://doi.org/10.3389/fcell.2014.000

Biochem From molecular signaling to modeling disease pathogenesis

The

Claire

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³ Department of Internal Medicine III, KH Elisabethinen Linz and Medical University of Vienna, Vienna, Austria

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ile Infants 60 Days or

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ical translation

in², Myrsini Kaforou²

.1111/imm.12841

Project development timeline

1. Development in time-course datasets of simple biological processes – circadian rhythm and *Plasmodium* intraerythrocytic life cycle stages --> My PhD thesis

2. Demonstration of use in time-course clinical datasets of whole-blood, predicting the future onset of:

a) Influenza A

b) SARS-CoV-2

3. Single time-point dataset: TB-IRIS

4. Demonstration of use in a large time-point clinical datasets: **PERFORM**



5. Reviewer demands: analysis of an additional SARS-CoV-2 and non-infectious disease datasets

Nat Comms
publication

6. Application to sepsis - ongoing work

7. Algorithm further development – PhD application in progress

For each dataset we demonstrated how the algorithms assumptions are met and then performed prediction analysis.

Results

The PERFORM study

“Febrile illness to Optimize Real-life Management across the European Union”

Subset with bacterial or viral diagnosis (n=399)
Recruited in the emergency department (ED) or PICU
Divided the patients by severity: mild, moderate and severe

Hospital recruited healthy controls also available

Definitions

Mild – sent **home** post-recruitment (recruitment was in the ED)
Moderate – sent to the **ward** (recruitment in the ED)
Severe – in the **PICU** (recruited in the PICU or ED)

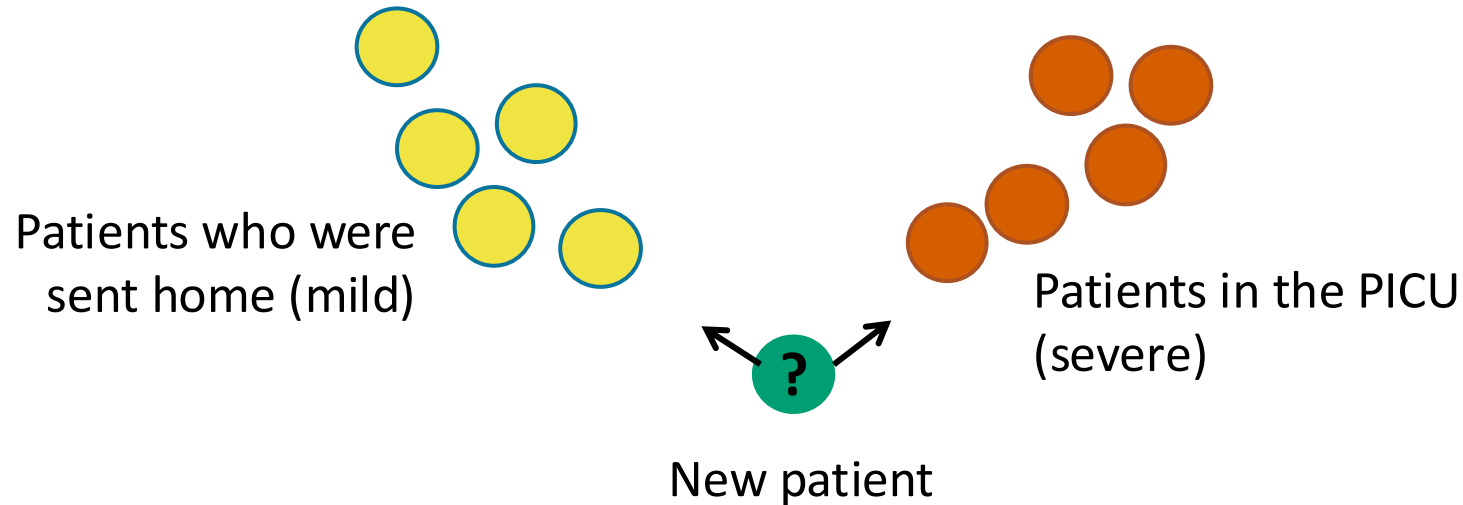
Diagnosis	Mild, N = 117		Moderate, N = 184		Severe, N = 98	
	Bacterial, N = 39 [†]	Viral, N = 78 [†]	Bacterial, N = 94 [†]	Viral, N = 90 [†]	Bacterial, N = 71 [†]	Viral, N = 27 [†]
Sex						
F	14 (36%)	44 (56%)	42 (45%)	39 (43%)	25 (35%)	8 (30%)
M	25 (64%)	34 (44%)	52 (55%)	51 (57%)	46 (65%)	19 (70%)
Sampling Location						
Emergency Department	39 (100%)	78 (100%)	94 (100%)	90 (100%)	6 (8.5%)	1 (3.7%)
Pediatric Intensive Care Unit	0 (0%)	0 (0%)	0 (0%)	0 (0%)	65 (92%)	26 (96%)
Death						
Yes	0 (0%)	0 (0%)	0 (0%)	0 (0%)	6 (8.5%)	1 (3.7%)
No	39 (100%)	77 (99%)	94 (100%)	90 (100%)	64 (90%)	26 (96%)
Unknown	0 (0%)	1 (1.3%)	0 (0%)	0 (0%)	1 (1.4%)	0 (0%)
Ethnicity						
North/Mid/East European	20 (51%)	33 (42%)	67 (71%)	56 (62%)	49 (69%)	16 (59%)
South Asian (Indian, Pakistani, Bangladeshi, Tamil)	1 (2.6%)	4 (5.1%)	1 (1.1%)	4 (4.4%)	0 (0%)	3 (11%)
South European	16 (41%)	26 (33%)	16 (17%)	12 (13%)	11 (15%)	2 (7.4%)
Other	1 (2.6%)	11 (14%)	9 (9.6%)	14 (16%)	9 (13%)	5 (19%)
Unknown	1 (2.6%)	4 (5.1%)	1 (1.1%)	4 (4.4%)	2 (2.8%)	1 (3.7%)
Ill Appearance						
Yes	8 (21%)	14 (18%)	41 (44%)	38 (42%)	55 (77%)	20 (74%)
No	25 (64%)	60 (77%)	43 (46%)	42 (47%)	5 (7.0%)	3 (11%)
Unknown	6 (15%)	4 (5.1%)	10 (11%)	10 (11%)	11 (15%)	4 (15%)
Sepsis						
Yes	4 (10%)	2 (2.6%)	21 (22%)	13 (14%)	56 (79%)	14 (52%)
No	35 (90%)	75 (96%)	72 (77%)	77 (86%)	15 (21%)	13 (48%)
Unknown	0 (0%)	1 (1.3%)	1 (1.1%)	0 (0%)	0 (0%)	0 (0%)
Diagnosis (Specific)						
Definite Viral	0 (0%)	46 (59%)	0 (0%)	72 (80%)	0 (0%)	15 (56%)
Probable Viral	0 (0%)	27 (35%)	0 (0%)	9 (10%)	0 (0%)	1 (3.7%)
Viral Syndrome	0 (0%)	5 (6.4%)	0 (0%)	9 (10%)	0 (0%)	11 (41%)
Definite Bacterial	12 (31%)	0 (0%)	61 (65%)	0 (0%)	48 (68%)	0 (0%)
Probable Bacterial	12 (31%)	0 (0%)	17 (18%)	0 (0%)	15 (21%)	0 (0%)
Bacterial Syndrome	15 (38%)	0 (0%)	16 (17%)	0 (0%)	8 (11%)	0 (0%)

[†] n (%)

Results

Proof of concept in the PERFORM study

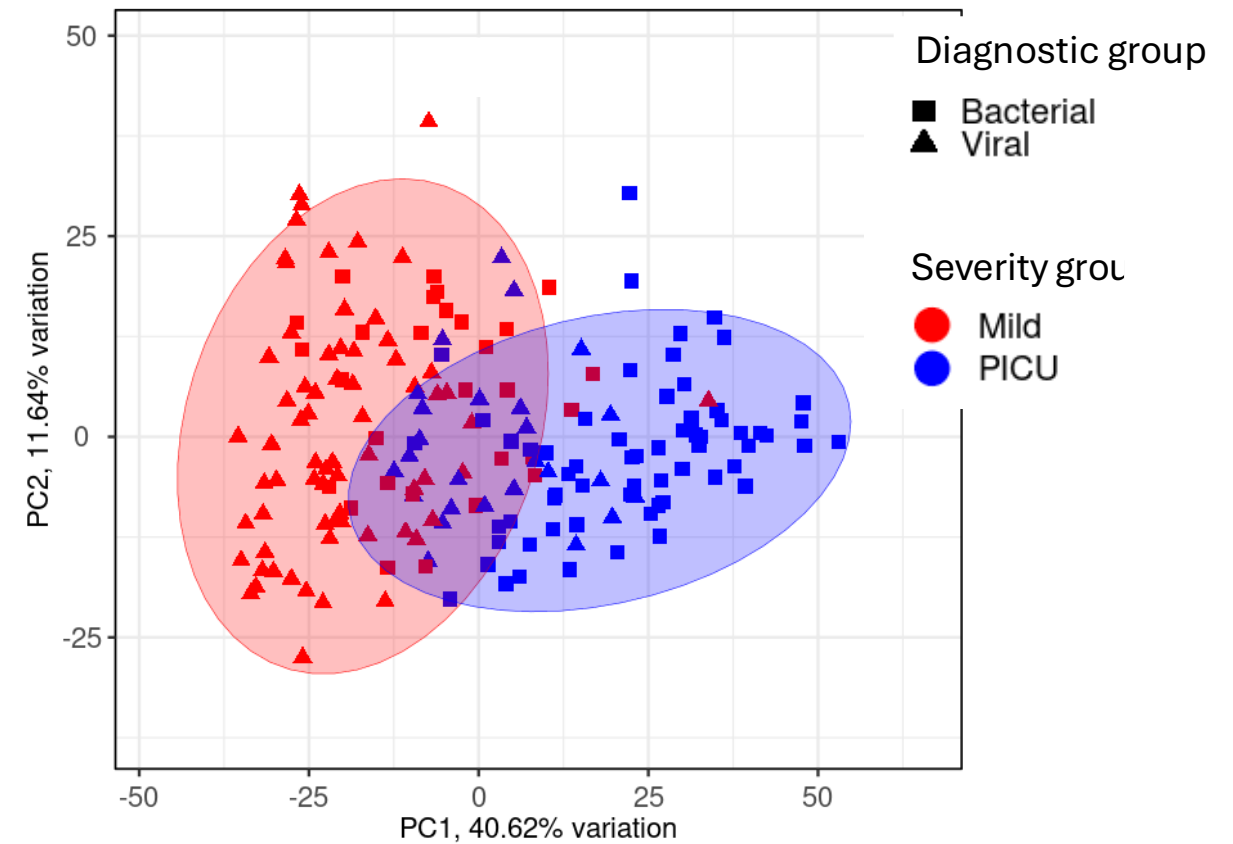
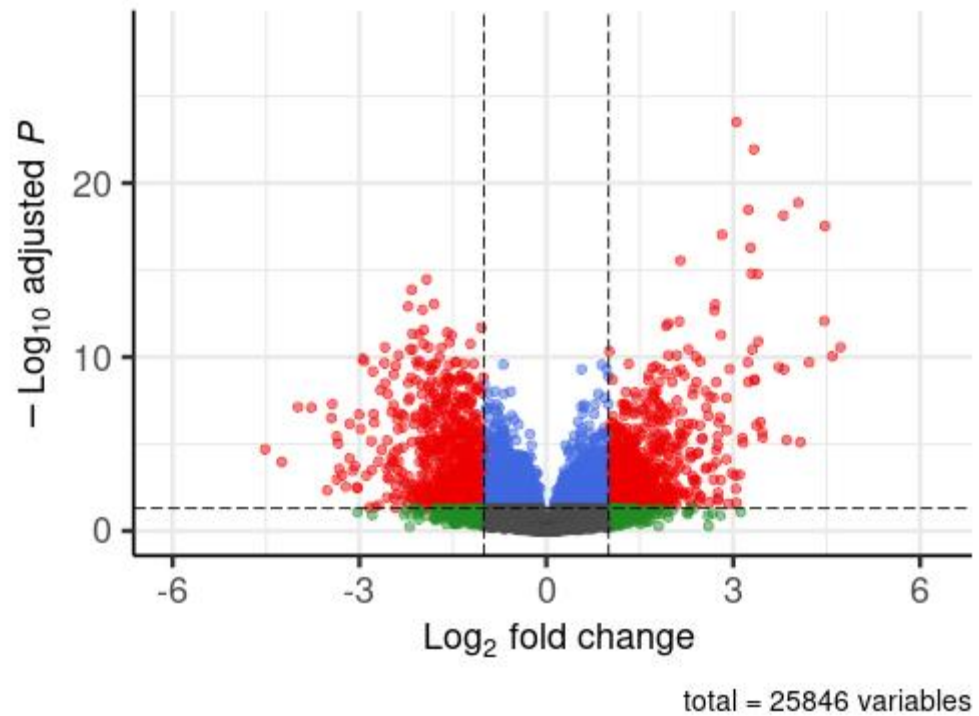
Reference groups: mild and severe groups recruited in the ED and PICU respectively



Test groups: moderate group recruited in the ED (7 of which later went to the PICU), 7 severe subjects recruited in the ED, 20 mild patients held back from the previous analysis

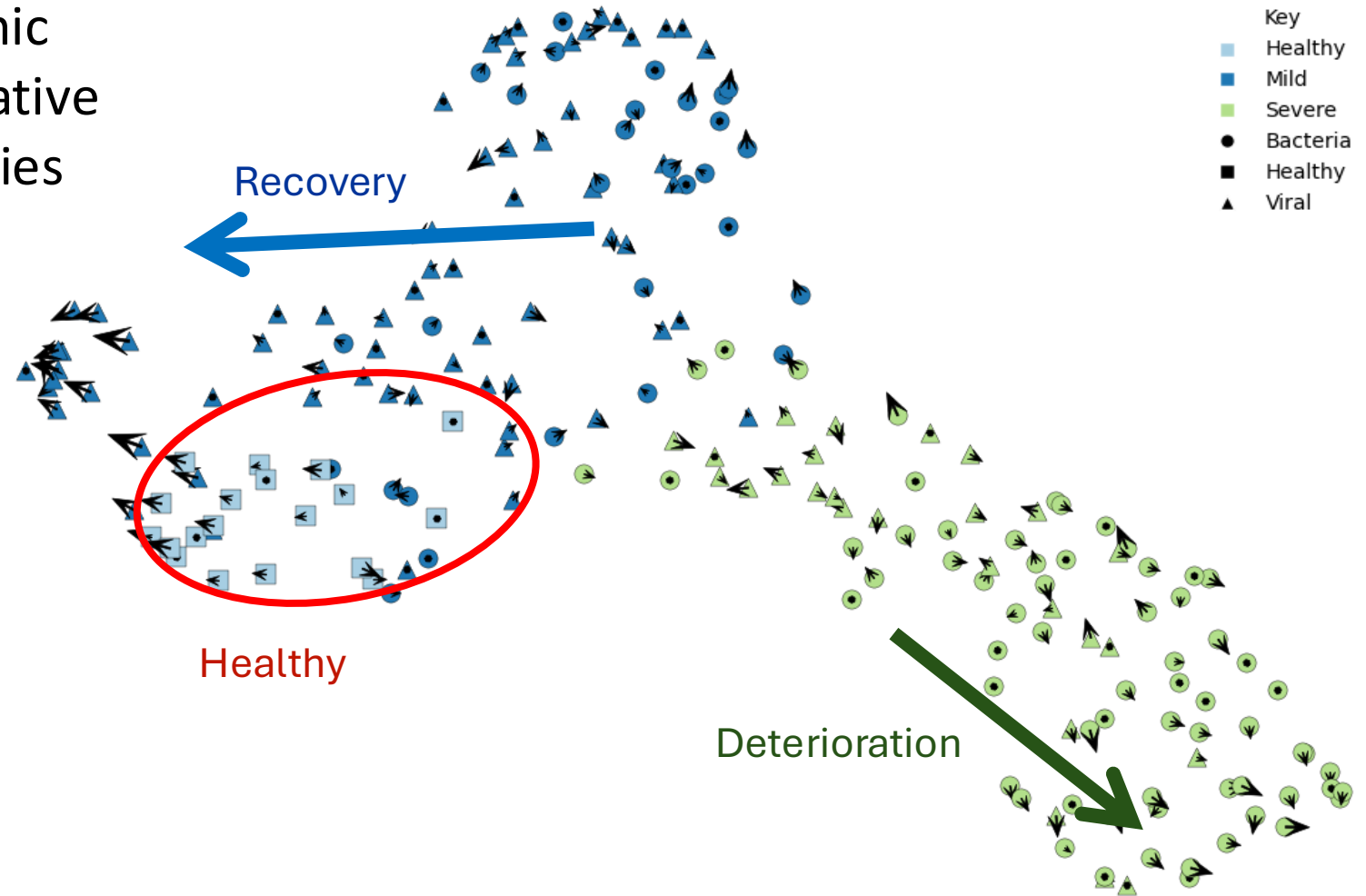
Results

Assumption 1: current transcriptomic state are representative of current disease states

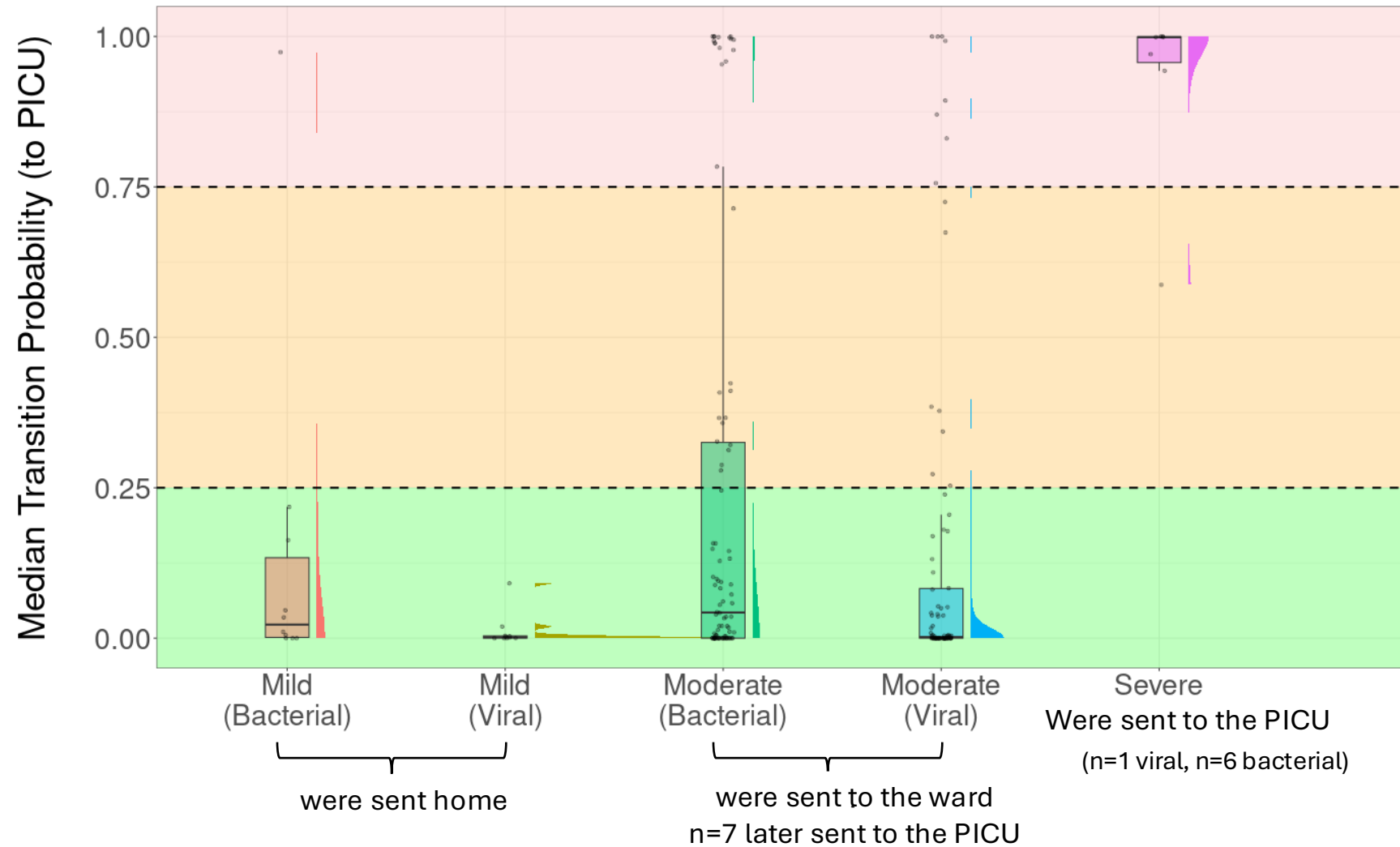


Results

Assumption 2: transcriptomic trajectories are representative of future disease trajectories



Results

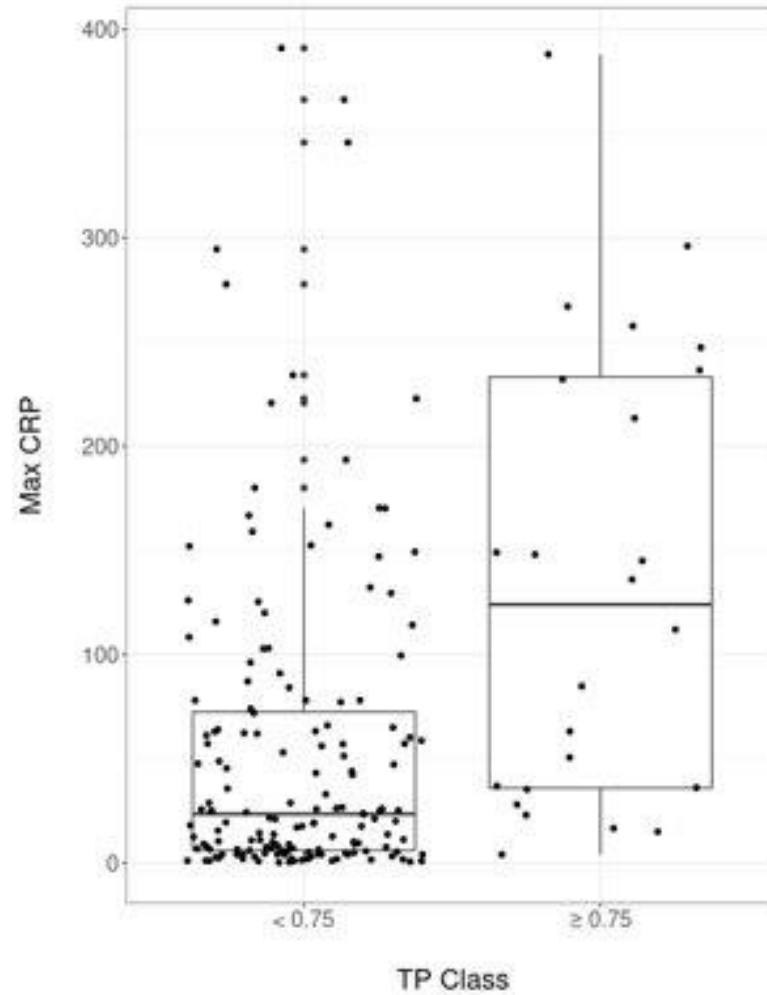


Results

ORs indicate that individuals in the more severe class (presence of sepsis, very urgent triage code etc.) are more likely to be in the high transition probability (TP, to the PICU) group.

Clinical Feature		TP ≥ 0.25 N (%)	TP < 0.25 N (%)	Odds Ratio	95% Confidence Interval	p-value
On-presentation						
Gender	Female	19 (44.19)	62 (43.97)	1.01	0.48, 2.12	1.00
	Male	24 (55.81)	79 (56.03)			
Triage Code	Very Urgent	27 (69.23)	79 (59.40)	1.53	0.68, 3.63	0.35
	Standard	12 (30.77)	54 (40.60)			
Ill Appearance	Yes	19 (51.35)	60 (47.24)	1.18	0.53, 2.62	0.71
	No	18 (48.65)	67 (52.76)			
Life Saving Intervention	Yes	4 (9.30)	13 (9.42)	1.00	0.22, 4.44	1.00
	No	39 (90.70)	125 (90.58)			
Phenotype	Bacterial	28 (65.12)	66 (46.81)	2.11	0.99, 4.65	0.038
	Viral	15 (34.88)	75 (53.19)			
Future Severity						
Surgery Performed	Yes	11 (25.58)	11 (7.80)	4.02	1.44, 11.28	0.0052
	No	32 (74.42)	130 (92.20)			
Transfer to the PICU						
	Yes	4 (9.30)	3 (2.13)	4.67	0.76, 33.18	0.053
	No	39 (90.70)	138 (97.87)			

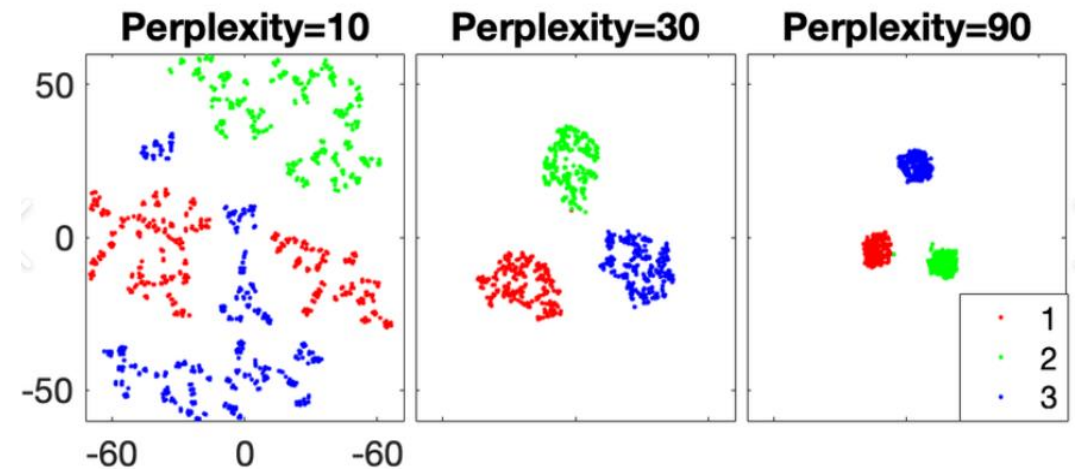
Results



Therefore, we **demonstrated proof of principle** that VeloCD can capture the relationship between predictions and future disease severity....

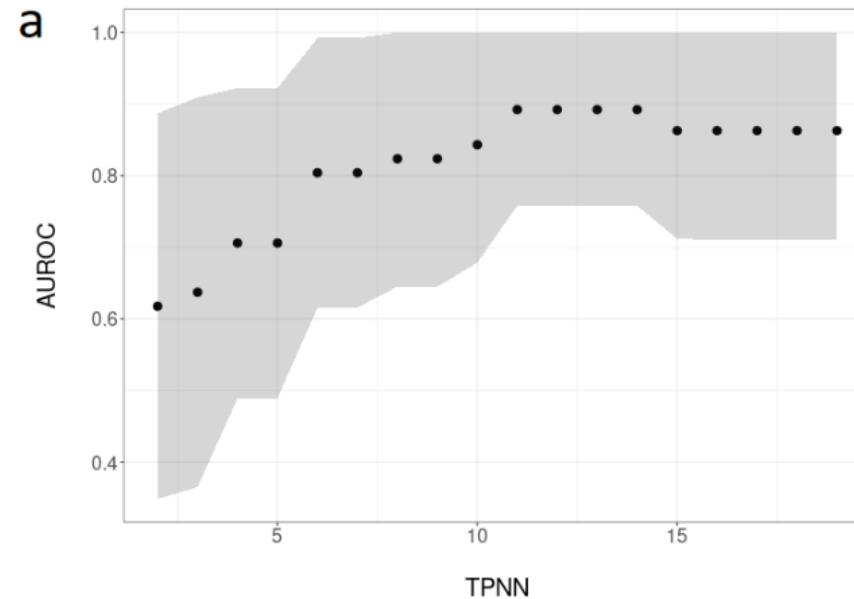
This is just a "proof of principle" of a complex algorithm

- What is proof of principle?
- What comes next?
- During algorithm development, I found many potential **pitfalls** areas of future improvement:
 - Choice of hyperparameters (NN, TPNN)
 - Choice of low-dimensional embedding methodology
 - Inherent stochasticity in tSNE and UMAP

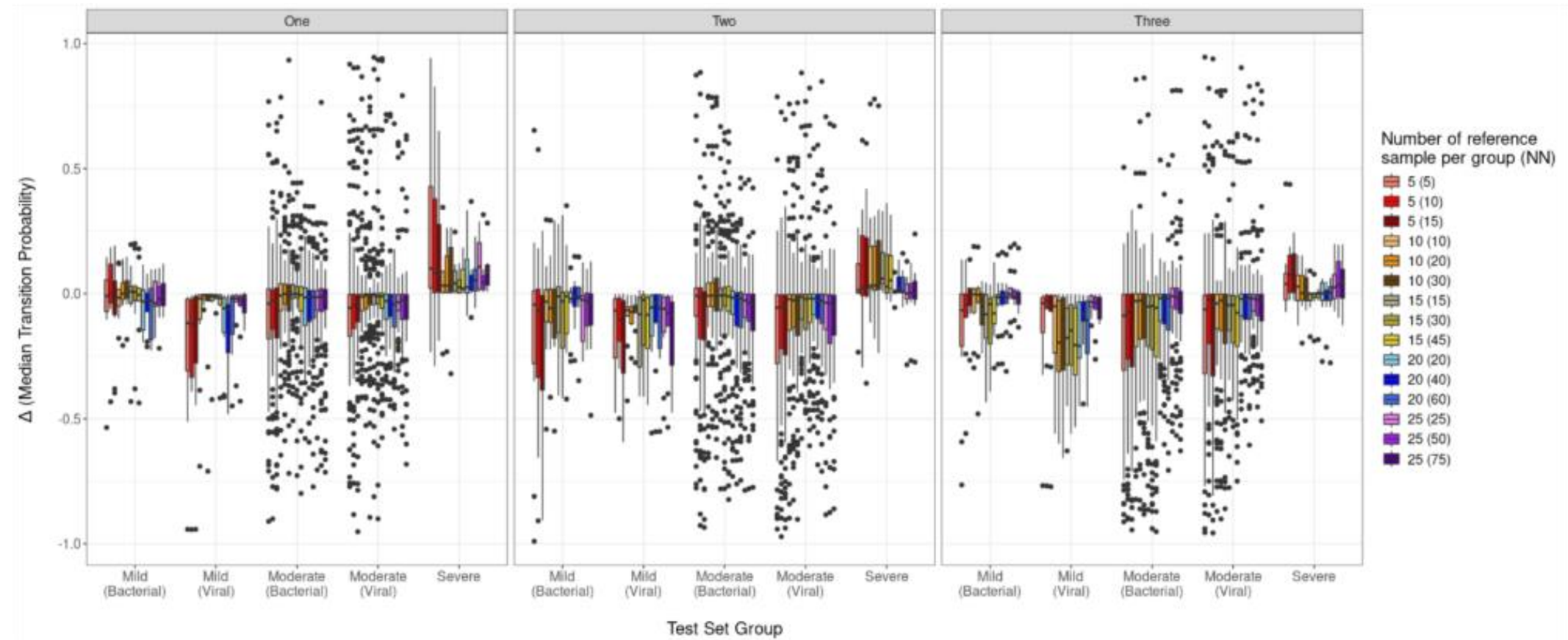


How should we handle this?

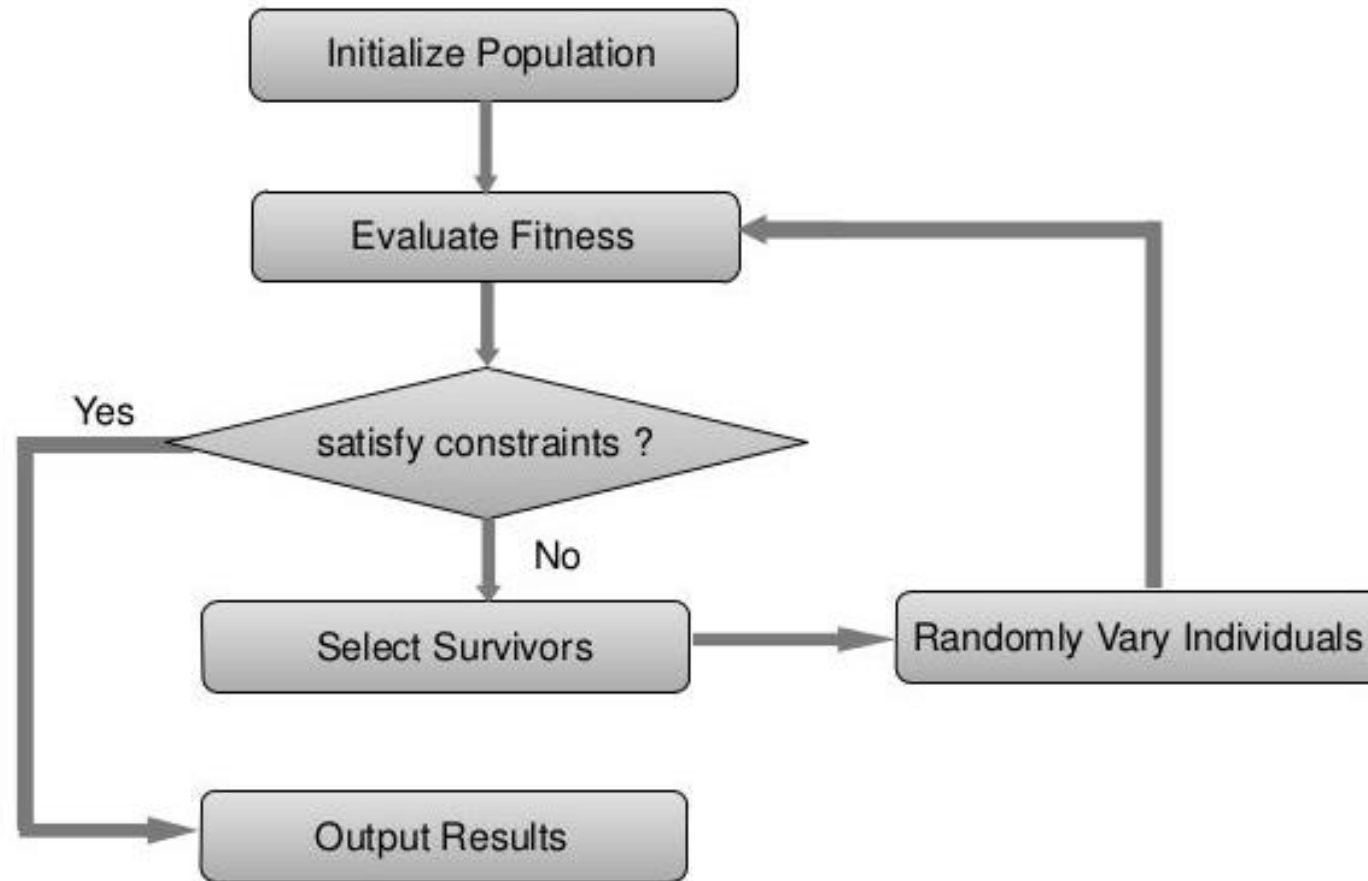
- Full transparency in the values used, the full range explored and an exploration of how the performance varies between hyperparameter values
- Some kind of median between tSNE and UMAP generations?



How many reference samples? what are the best reference samples?



Genetic algorithms to optimize hyperparameter selection



 **DrClaireDunican / VeloCD** Public

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 **1 Branch**

 **0 Tags**

<> Code ▾

 **DrClaireDunican** Update CalculateSplicedUnspliced.R ...

4793f32 · 2 weeks ago  **53 Commits**

 ExampleFiles	Add files via upload	last year
 Preprocessing	Update CalculateSplicedUnspliced.R	2 weeks ago
 Analyse Prediction Results.R	Update Analyse Prediction Results.R	4 months ago
 ConfidenceIntervalsAcrossTPNN.R	Core source code for VeloCD	2 years ago
 InternalBootStrappingTransitionProbabilities...	Core source code for VeloCD	2 years ago
 LICENSE	Initial commit	2 years ago
 README.md	Update README.md	last year
 RNA_Velocity_Main.py	Core source code for VeloCD	2 years ago
 RunRNAVelocity.sh	Core source code for VeloCD	2 years ago

About

No description, website, or topics provided.

 Readme

 Apache-2.0 license

 Activity

 **0 stars**

 **1 watching**

 **1 fork**

Report repository

Releases

No releases published

Packages

No packages published

The present...

After 7.5 long years the preprint is out.....and it has been accepted for publication at Nature Communications



Thank you for submitting your manuscript "Predicting trajectories of illness using RNA velocity of whole blood" to **Nature Communications**. I am delighted to say that we are happy, in principle, to publish it under an open access license.

The future...

Immediate: further work on its application to sepsis, optimization of predictions using machine learning (autoencoders and genetic algorithms)

Medium term: moving past RNA-Seq data to point of care unspliced and spliced transcript measurements...

Long term goal is to adapt this tool into something that could impact clinical care decision making...

Take home messages

1. Knowing when to stop – when is the project no longer proof of principle? What are the next steps? When should it be handed over to others to progress?
2. Acknowledging the limitations of the work – how to frame, report upon and improve



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Acknowledgements

Core Team: Myrsini Kaforou, Mauricio Barahona, Aubrey Cunningham.

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The PaedsID Team (especially the malaria and Bioinformatics subgroups)

Ashleigh Green

